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Studierichting: Informatica
Oefeningenreeks 4A nr. 52.2

$$\int \sin^2 x \cos^4 x dx = \int (\sin x \cos x)^2 \cos^2 x dx$$
$$= \int \left(\frac{1}{2}\sin 2x\right)^2 \cos^2 x dx = \frac{1}{4}\int \sin^2 2x \cos^2 x dx$$
$$= \frac{1}{16}\int (1-\cos 4x)(1+\cos 2x) dx = \frac{1}{16}\int (1+\cos 2x-\cos 4x-\cos 4x \cos 2x) dx$$
$$= \frac{x}{16} + \frac{1}{32}\int \cos 2x dx - \frac{1}{64}\sin 4x - \frac{1}{32}\int \cos 6x dx$$
$$= \frac{x}{16} + \frac{1}{64}\sin 2x - \frac{1}{64}\sin 4x - \frac{1}{192}\sin 6x + c$$

References